

CSE260 (QUIZ - 2 (33))

Summer2025

Fullmark: 15

ID: _____

Name: _____

Section: _____

Q1 : Implement XOR gate using NAND gate only .

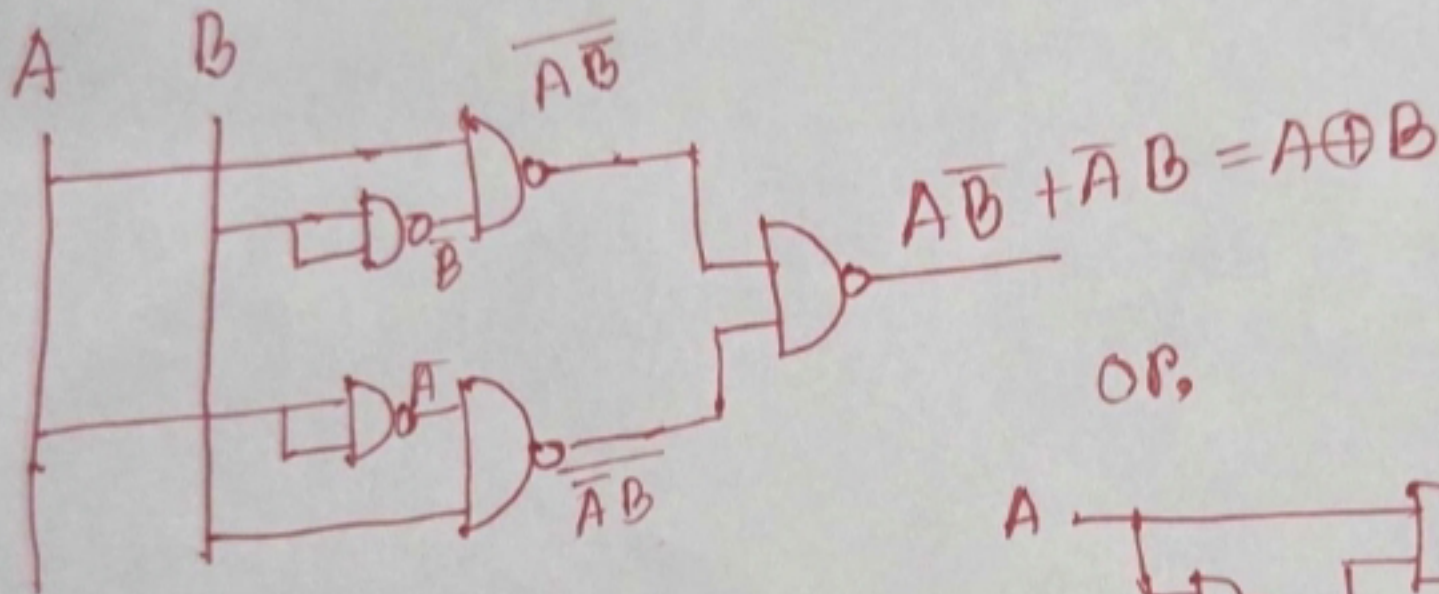
[8]

Q2 : FIND canonical POS format for $f(w,x,y,z) = x+yz$

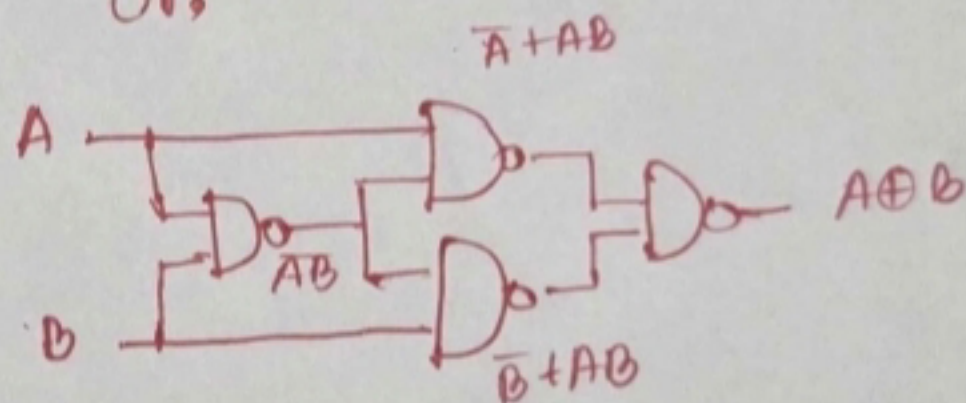
[7]

Q3 : What is MINTERM? [Bonus : 1]

Q1: $XOR \rightarrow AB + \bar{A}B$



OR,



Q2 : $x + yz$
 $= (x+y)(x+z)$
 $= (x+y+z\bar{z}+w\bar{w})(x+z+y\bar{y}+w\bar{w})$
 $= (x+y+\bar{z}+\bar{w})(x+y+\bar{z}+w)(x+y+z+\bar{w})(x+y+z+w)$
 $= (x+z+\bar{y}+\bar{w})(x+z+\bar{y}+w)(x+z+y+\bar{w})(x+z+y+w)$
 $= \prod (0, 1, 2, 8, 9, 10)$